

# Data Modeling Intern — Assignment

**Estimated Time:** 4–6 hours

**Submission Format:** A single PDF or markdown document with all your answers, schemas, and reasoning clearly written out.

**Tools Allowed:** Any — including AI tools. However, be aware that evaluation is based on your reasoning, not just your output.

## A Note Before You Begin

This assignment is intentionally open-ended in places. There are no perfect answers. What we are evaluating is **how you think** — the decisions you make, the trade-offs you recognize, the assumptions you call out, and the questions you would ask before designing anything.

A well-reasoned "imperfect" design with clear justification will always score higher than a perfect-looking design with no explanation.

## The Scenario

You are joining the team building **FlowDesk** — an internal tool that helps organizations manage multi-step document approval workflows.

Here is a brief product description written by the product manager:

*"FlowDesk allows teams to submit documents for approval. A document goes through multiple reviewers in a sequence. Each reviewer can approve or reject. If rejected, the submitter can revise and resubmit. Admins can configure how many reviewers a document needs and who they are. We also want to track the full history of what happened to every document."*

This is all the context you have. The engineering has not started yet. You are the first person touching the data layer.

## Part 1 — Requirements Analysis (Before You Design Anything)

**Do not jump into designing yet.**

1. List **at least 8 specific questions** you would ask the product manager before starting any design work. These should be questions whose answers would meaningfully change your data model.

2. For each question, write **one sentence explaining why that answer matters** to your design.
3. After listing your questions, write down **the assumptions you are making** in order to proceed with the rest of this assignment. Be explicit — "I am assuming X because Y was not specified."

**Evaluation note:** This section reveals whether you can spot ambiguity. A candidate who skips straight to designing without questioning the requirements is a red flag in any engineering team.

## Part 2 — Database Schema Design

Based on your assumptions from Part 1, design a relational database schema for FlowDesk.

**Your submission must include:**

1. All tables with column names, data types, and constraints (primary keys, foreign keys, nullability, unique constraints).
2. A brief description of what each table represents.
3. An ER diagram or a clearly written description of how tables relate to each other.

**You must also answer the following questions about your own schema:**

- Why did you choose to structure the reviewer sequence the way you did? What alternatives did you consider?
- Where did you choose to store document history/audit trail, and why? What are the trade-offs of your approach?
- Is there anything in your schema that you are not fully confident about? If yes, what is it and what would help you decide?

## Part 3 — API Contract Design

Design the REST API for the following three operations, for each, provide the full request and response structure:

1. **Submit a new document for approval**
2. **A reviewer takes an action on a document (approve or reject)**
3. **Fetch the full history/timeline of a document**

For each API:

- Define the HTTP method and endpoint path.

- Define the complete request body (if any) with field names, types, and whether they are required or optional.
- Define the success response body.
- Define at least two possible error responses with appropriate HTTP status codes.
- Write 2–3 lines explaining a key design decision you made for that particular endpoint.

## Part 4 — Spot the Problems

Below is a schema that a junior developer proposed for storing the reviewer sequence of a document. **It has multiple problems.** Your job is to find them.

**Table:** document\_reviewers

| Column          | Type     |
|-----------------|----------|
| id              | INT (PK) |
| document_id     | INT      |
| reviewer_1_id   | INT      |
| reviewer_2_id   | INT      |
| reviewer_3_id   | INT      |
| reviewer_1_done | BOOLEAN  |
| reviewer_2_done | BOOLEAN  |
| reviewer_3_done | BOOLEAN  |
| all_approved    | BOOLEAN  |

1. List every problem you can identify with this schema. Be specific — explain *why* each thing is a problem, not just *what* it is.
2. For each problem, explain what breaks or becomes difficult because of it.
3. Show how your schema from Part 2 addresses these problems.

## Part 5 — The Curveball

After you have completed Parts 2 and 3, the product manager comes back with this new requirement:

*"We also want to support parallel approvals — meaning sometimes all reviewers should review at the same time, and the document is approved only when all of them approve. Sometimes it should still be sequential. This should be configurable per workflow."*

### Answer the following:

1. Does this new requirement break, conflict with, or complicate anything in your existing schema or API design? Be honest — if it does, say so clearly.
2. What changes would you make to your schema to support this?
3. What changes would you make to your API contracts?
4. Is there anything in this new requirement that is still ambiguous to you? What additional questions would you ask?

**Evaluation note:** We are not looking for a perfect redesign here. We are looking for whether you can recognize impact, adapt thoughtfully, and stay honest about what you don't know.

## Submission Checklist

Before submitting, make sure your document includes:

- Part 1: At least 8 questions with reasoning and your explicit assumptions
- Part 2: Full schema with types, constraints and written justifications
- Part 3: Request/response definitions for all 3 APIs, with error cases and design notes
- Part 4: A clear list of problems with the given schema and explanations
- Part 5: Impact analysis, changes and remaining questions

## How This Will Be Evaluated

| What We Look At               | What We're Testing                                     |
|-------------------------------|--------------------------------------------------------|
| Questions you asked in Part 1 | Ability to spot ambiguity before writing code          |
| Justifications in Part 2 & 3  | Depth of reasoning, not just correctness               |
| How you handle Part 4         | Ability to critically review existing designs          |
| Honesty in Part 5             | Intellectual integrity and adaptability                |
| Overall clarity               | Communication — can you explain your thinking clearly? |

We will discuss your submission during the interview call. Be prepared to walk us through your decisions and answer "why did you do it this way?" for any part of your design.

*Good luck. Take your time. Think out loud on paper.*